Implement the cannon algorithm on a 2D distribution of data

- If you missed/want to review the in class explanation see this video:
  - https://www.youtube.com/watch?v=6r6uPImUUBQ
- The following slides are as a reference to use the MPI cartesian grid related functions
- Find out the highest real peak your MM code can deliver on the DCGP partition either using all_gather or cannon!
- Come at the exam with a performance analysis and plots that show how your code/s scale with respect to the increasing of NPEs

3D cartesian topology

```
MPI_Cart_create (MPI_COMM_WORLD, 3, dims, periods, 0, &MPI_COMM_CART);


/* Build the sub-communicators along X and Y */
coords[0] = 1; coords[1] = 0; coords[2] = 0;
MPI_Cart_sub (MPI_COMM_CART, coords, &MPI_COMM_ALONG_X);
coords[0] = 0; coords[1] = 1; coords[2] = 0;
MPI_Cart_sub (MPI_COMM_CART, coords, &MPI_COMM_ALONG_Y);
coords[0] = 0; coords[1] = 0; coords[2] = 1;
MPI_Cart_sub (MPI_COMM_CART, coords, &MPI_COMM_ALONG_Z);

/*! logical mapping of neighbour tasks ONLY for cartesian grid (using mpi functions) */
MPI_Cart_shift (MPI_COMM_ALONG_X, 0, 1, &pxm, &pxp);
MPI_Cart_shift (MPI_COMM_ALONG_Y, 0, 1, &pym, &pyp);
MPI_Cart_shift (MPI_COMM_ALONG_Z, 0, 1, &pzm, &pzp);
```

3D cartesian topology: comm pattern

```
MPI_Sendrecv( field + NZP2*NYP2*NX + NZP2+1, 1, MPI_X_RhoPlane, pxp, 20,
              field + NZP2+1 ,  1, MPI_X_RhoPlane, pxm, 20, MPI_COMM_ALONG_X, &status1);
MPI_Sendrecv( field + NZP2*NYP2 + NZP2+1, 1, MPI_X_RhoPlane, pxm, 21,
              field + NZP2*NYP2*(NX+1) + NZP2+1 , 1, MPI_X_tytpe, pxp, 21, MPI_COMM_ALONG_X, &status1);


MPI_Sendrecv( field + NZP2*(NY) + 1, 1, MPI_Y_RhoPlane, pyp, 22,
              field + 1,  1, MPI_Y_RhoPlane, pym, 22, MPI_COMM_ALONG_Y, &status1);
MPI_Sendrecv( field + NZP2 + 1, 1, MPI_Y_RhoPlane, pym, 23,
              field + NZP2*(NY+1) + 1, 1, MPI_Y_type, pyp, 23, MPI_COMM_ALONG_Y, &status1);


MPI_Sendrecv( field + NZ,  1, MPI_Z_type, pzp, 24,
              field,  1, MPI_Z_type, pzm, 24, MPI_COMM_ALONG_Z, &status1);
MPI_Sendrecv( field + 1, 1, MPI_Z_RhoPlane, pzm, 25,
              field + NZ+1, 1, MPI_Z_RhoPlane, pzp, 25, MPI_COMM_ALONG_Z, &status1);
```